

Yury Shirokov

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SUMMARY

Rising senior at UC Berkeley (Economics, Data Science minor) with hands-on experience building full-stack products, ML systems, and AI-integrated applications. Focused on shipping real, production-grade software end to end — from architecture to AI integration to security.

EDUCATION

University of California, Berkeley

B.A. Economics, Minor in Data Science

Expected May 2027

PROJECTS

PerfinLab — Full-Stack Personal Finance SaaS (React 19, Flask, Firebase, Claude API, Stripe)

- Shipped a production SaaS app (v1.7.0, 21K+ LOC) on a React 19 + Flask + Firebase Cloud Functions architecture, with Stripe subscription billing and CI/CD via GitHub Actions.
- Integrated Claude Sonnet 4.6 across 6 features, including a streaming agentic advisor and AI document extraction from bank statements (PDF/CSV/image); implemented TOTP 2FA and Fernet-encrypted Plaid tokens.

Vanguard RL — Multiplayer Environment AI Agent (Python, C#, PyTorch, Stable-Baselines3)

- Built a high-performance cross-process bridge between a C# game engine plugin and a PyTorch RL core using UDP-based synchronization for 10Hz frame-locked updates, with a custom ResNet-18 vision backbone for real-time spatial data.
- Used SubprocVecEnv to bypass Python's GIL, running an 8-way parallel bot swarm that reached 200 steps/sec and saturated GPU compute across 1.6M+ timestep training runs.

Quantbot — Hybrid Options Trading Engine (Python, FastAPI, React, Gymnasium, yfinance)

- Engineered a decoupled Python/FastAPI backend with a React telemetry dashboard, isolating high-frequency trading logic from I/O-heavy monitoring.
- Built a sub-millisecond technical-analysis engine in pure Pandas (EMAs, Bollinger Bands, Stochastic Oscillators) and trained a PPO agent to generate signals in the QQQ options market.

Data Analytics: E-commerce Funnel & A/B Testing (Google BigQuery, Python)

- Analyzed 12M+ rows of session data in BigQuery SQL to isolate a 56.2% drop-off at product-view → add-to-cart, the highest-leverage point in the funnel.
- Designed and validated an A/B test (power analysis, two-proportion z-tests, bootstrapped 95% CIs) confirming a statistically significant 5% lift in mobile checkout conversion.

EXPERIENCE

FloodDry Water Damage Restoration (San Jose, CA) — CRM Administrator & Automation Engineer *Apr 2026 – Present*

- Implemented a CRM and migrated the company's phone system (Verizon → Twilio) with full call tracking for a small business.
- Built an agentic AI workflow automating lead generation, job scheduling, and subcontractor communication end to end.

Southwest YMCA (Saratoga, CA) — Personal Trainer / Fitness Attendant / Group Exercise Instructor *Aug 2023 – Present*

- Supported 500+ members/day in a high-traffic facility; led group classes of 15–30 participants.

SKILLS

Languages: Python, JavaScript (React 19), SQL, C++, C#, R

Frameworks/Tools: Flask, Firebase (Firestore/Auth/Hosting), PyTorch, Stable-Baselines3, pandas, scikit-learn, Tailwind CSS, GitHub Actions

APIs/Platforms: Claude API, Plaid API, Stripe, Google Gemini API, Firebase

Languages Spoken: Russian (native)